External Interrupts & Events (EXTI) Overview

Overview & Implementation Steps

External Interrupts (EXTI) Overview

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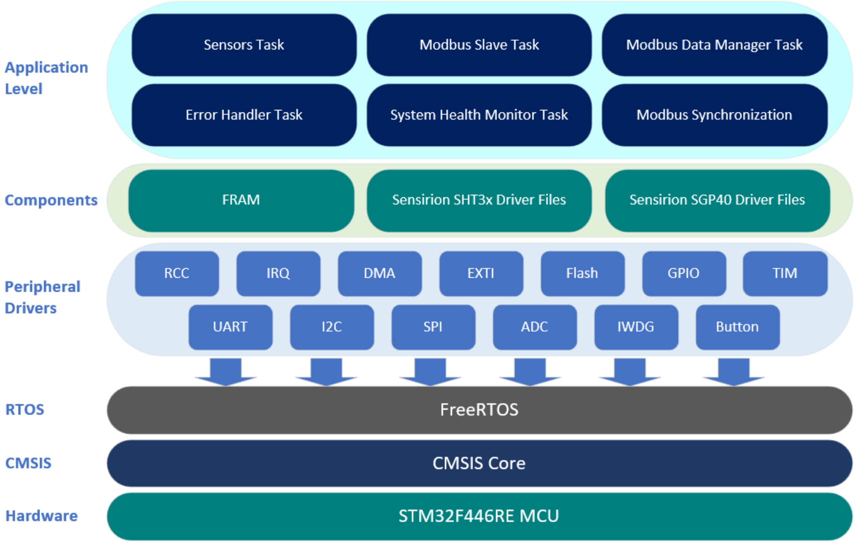
External Interrupts (EXTI) Overview

- Definition: The **External Interrupt / Event Controller** (**EXTI**) manages external interrupt lines, allowing peripherals to signal the microcontroller to execute an interrupt service routine (ISR).

- Purpose & Functionality:
  - **Event Detection**: Detects external events (e.g., button presses) and triggers corresponding interrupt handler.

- Project Significance:
  - **User Button Acknowledgement**: During startup, the system checks for a previous IWDG reset and waits for user acknowledgment via the button press to continue.



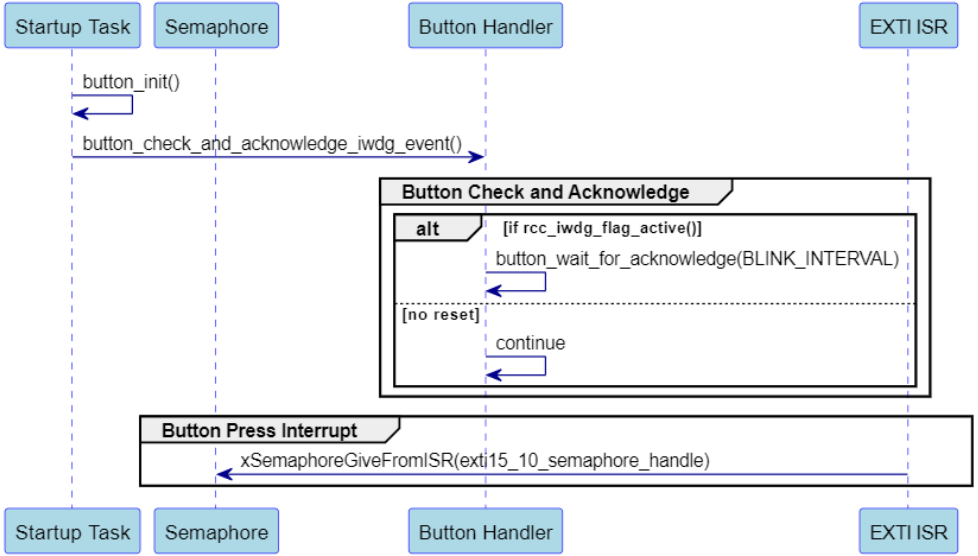
EXTI Use in the Project

Functionality & Purpose

- **EXTI** configured to be triggered by USER button press to acknowledge the IWDG reset.

Key Points

- **Initialized for USER Button**: Configures EXTI line with falling edge trigger interrupt.
- **Button Handler**: Uses a Semaphore to synchronize user acknowledgment.



- **EXTI ISR**: EXTI15_10_IRQHandler Gives Semaphore upon button press allowing the system to proceed.

EXTI Basic Use

EXTI Line Mapping

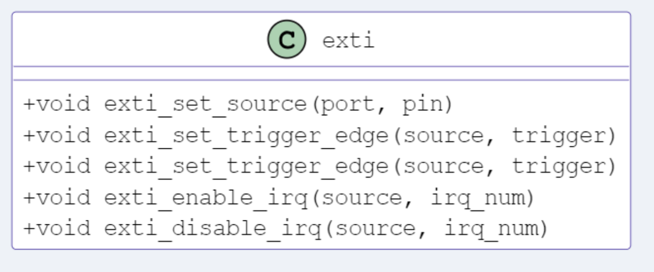
- **EXTI0-EXTI15:** GPIOs are connected to 16 EXTI lines *(page 246 ref. manual)*.

Key Points

- **Configure EXTI Line Source**: Set the EXTI line source for EXTI0-EXTI15 by specifying the GPIO Pin in the SYSCFG->EXTICR[0…3] *(page 197 ref. manual)*.
- **Set Trigger Edge**: Configure the edge (Rising, Falling, or Both) on which the interrupt triggers using EXTI->RTSR & EXTI->FTSR registers *(page 248 ref. manual)*.
- **Enable EXTI Interrupt**: Unmask the specified EXTI interrupt, enabling it to trigger using EXTI->IMR *(page 247 ref. manual)* and enable it by calling NVIC_EnableIRQ().
- **Disable EXTI Interrupt**: Masks the specified EXTI interrupt, preventing it from triggering using EXTI->IMR and disable it by calling NVIC_DisableIRQ().

EXTI Driver Overview

Functions



- **Set Source**: Configures which GPIO pin/port acts as interrupt source by updating SYSCFG->EXTICR[0...3] using port & pin enumerations.
- **Set Trigger Edge**: Sets the edge sensitivity (rising, falling, or both) for the specified EXTI source by using source & trigger enumerations.
- **Enable IRQ**: Unmasks the EXTI interrupt in the interrupt mask register (IMR) using the source enumeration and IRQn_Type and enables it through the NVIC by calling NVIC_EnableIRQ().
- **Disable IRQ**: Masks the EXTI interrupt in the IMR and disables it through the NVIC using the source enumeration and IRQn_Type and disables it through the NVIC by calling NVIC_EnableIRQ().

Practical Implementation Steps (Header File)

Key Points

- **PORT Enumeration**: Represents GPIO ports (A to H) used to set the PORTA-H values in the SYSCFG->EXTICR[0...3] register.
- **EXTI Source Enumeration**: Defines interrupt lines (0-15) and system-specific handlers for external interrupts.
- **Trigger Edge Enumeration**: Specifies the edge sensitivity (Rising, Falling, Both) for interrupts in EXTI->RTSR and EXTI->FTSR.
- **Public Functions**: Declare interface functions to configure and control EXTI lines and interrupts in the application code.

Importance

- **Interface Provision**: Provides a clear interface for the source file, ensuring modularity and ease of use.

Practical Implementation Steps (Source File)

Key Points

- **Define Public Functions**
  - void exti_set_source(exti_port_e port, gpio_num_e pin)
  - void exti_set_trigger_edge(exti_source_e source, exti_trigger_e trigger)
  - void exti_enable_irq(exti_source_e source, IRQn_Type irq_num)
  - void exti_disable_irq(exti_source_e source, IRQn_Type irq_num)

Importance

- Provides a comprehensive set of functions to manage EXTI configurations, including selecting sources, setting trigger edges, and enabling/disabling their related interrupts.

Next: EXTI Header File